

Reviewer Analysis & Consensus

PLOS Computational Biology | arXiv:2603.12416

This document synthesizes the feedback from 9 reviewers regarding the manuscript extending the Assembly Calculus (AC) model with E%-WTA dynamics and feedforward inhibition. The feedback is organized by major thematic areas, followed by a consensus map detailing intersecting critiques.

1. Model Comparison & Experimental Design

Reviewers widely noted that the comparison between the proposed E%-WTA model and the original AC model is confounded by simultaneous parameter changes and differing criteria.

- **Confounded Variables:** Multiple reviewers highlighted that the proposed model alters the selection rule (k-WTA to E%-WTA), adds inhibitory synapses, alters connection probabilities (p_s from 0.1 to 0.5), and uses stricter formation criteria all at once. ([Rev 1](#) [Rev 4](#) [Rev 5](#) [Rev 9](#) [PR 2](#))
- **Ablation Studies Required:** There is a strong consensus that a factorial ablation study is necessary. The new stopping criteria must be applied to the baseline AC model to see if the improvements are due to the stopping rule or the E%-WTA mechanism. ([Rev 1](#) [Rev 5](#) [Rev 6](#))
- **Unfair Baseline:** Comparing models using different stopping conditions (one requiring a fixed point, the other not) creates a tautological advantage for the proposed model. ([Rev 5](#) [Rev 9](#) [PR 2](#))

2. Biological Plausibility

While the paper aims to increase biological realism, several reviewers pointed out structural decisions that contradict biological observations.

- **Network Sparsity:** The connection probability ($p = 0.5$) is far higher than the biological estimates of ~ 0.01 – 0.1 for cortical pyramidal cells. The model reportedly fails at lower sparsities, which undermines the biological plausibility claim. ([Rev 4](#) [Rev 5](#) [Rev 9](#))
- **Dale's Law & Inhibition:** The implementation of inhibition (assigning negative weights to random synapses) does not align with Dale's law or the physical separation of excitatory and inhibitory neurons. The E/I ratio cited usually refers to cell counts, not synapse probabilities. ([Rev 6](#) [Rev 9](#))
- **Inhibitory Plasticity:** Applying the same multiplicative Hebbian rule to both excitatory and inhibitory synapses (creating anti-Hebbian effects) is questioned. Reviewers recommend standard homeostatic plasticity rules for inhibition. ([Rev 1](#) [Rev 5](#) [Rev 6](#))

3. Missing Functional & Cognitive Tests

Reviewers felt that crucial tests to prove the cognitive capabilities of the new assemblies were omitted.

- **Pattern Completion & Partial Cues:** Testing retrieval by presenting the exact same, complete stimulus is considered a trivial test. The model must be tested on degraded or partial cues to prove true attractor stability. ([Rev 1](#) [Rev 5](#) [PR 2](#))
- **Sequence Memory & Assembly Operations:** The original AC paper defined operations like 'project' and 'merge'. Reviewers requested verification that the new E%-WTA model can still support these compositional operations and sequence memory. ([Rev 1](#) [Rev 6](#) [Rev 8](#))
- **Storage Capacity:** The model's capacity limit and susceptibility to catastrophic forgetting when storing multiple assemblies must be quantified. ([Rev 5](#) [PR 2](#))

4. Metrics and Justification of Parameters

Several computational metrics and constraints introduced by the authors need robust mathematical or biological justification.

- **Overlap Normalization:** Raw overlap is an unfair metric for orthogonalization because E%-WTA creates smaller assemblies than k-WTA. The overlap must be normalized by assembly size (e.g., Jaccard index or cosine similarity). ([Rev 5](#) [Rev 8](#) [Rev 9](#))
- **Synaptic Density Metric:** Because non-zero synapses cannot go to zero and zero synapses stay zero under the multiplicative rule, counting the physical *number* of synapses is flawed. Weight-based density should be reported instead. ([Rev 2](#) [Rev 5](#))
- **Arbitrary Size Constraint:** The hard-coded minimum assembly size ($|A| \geq 6$) prevents degenerate solutions (1 neuron per stimulus) but is fundamentally arbitrary. The model should ideally prevent degenerate solutions organically. ([Rev 5](#) [Rev 9](#) [PR 2](#))
- **Synaptic Normalization:** Unbounded multiplicative growth of weights can destabilize the network; it must be clarified whether weights are normalized (as in the original AC). ([Rev 5](#) [Rev 6](#))

5. Presentation and Manuscript Structure

A consistent critique was that the manuscript reads as overly descriptive and lacks narrative flow and mechanistic insight.

- **Mechanistic vs. Descriptive:** The Results section describes *what* happens (performance improvements) without explaining *why* or *how* the dynamics (like gamma oscillations) cause these improvements. Analytical explanations are strongly requested. ([Rev 2](#) [Rev 4](#) [Rev 8](#))
- **Structural Edits:** The "Limitations" section is currently placed in Results but belongs in the Discussion. The Discussion needs rewriting to better contextualize findings against existing literature. ([Rev 4](#) [Rev 9](#))
- **Figures and Visibility:** Labels, legends, and ticks on nearly all figures are too small to read. Error bars are missing on key plots (e.g., Fig 4A/B), and significance markers should be added directly to the graphs. ([Rev 1](#) [Rev 2](#) [Rev 4](#) [Rev 8](#))

6. Consensus Map of Critical Intersections

The following table identifies the most critical overlapping critiques raised by multiple reviewers, highlighting areas that should be prioritized during manuscript revision.

Reviewer Consensus	Core Critique	Impact Area
Reviewer 1 Reviewer 4 Reviewer 5 Reviewer 6 Reviewer 9 PR_2	Confounded Model Comparison The comparison to the original AC model is invalid because multiple variables (stopping criteria, connection probability p_s , selection mechanism, inhibitory synapses) were changed simultaneously. Factorial ablations are mandatory.	Experimental Design
Reviewer 4 Reviewer 5 Reviewer 6 Reviewer 9	Biological Implausibility Features intended to increase biological realism are fundamentally unbiological. Specifically: network sparsity ($p = 0.5$) is far too dense, inhibition violates Dale's law, and E/I ratio application is flawed.	Biological Plausibility
Reviewer 5 Reviewer 8 Reviewer 9	Invalid Overlap Metric Comparing absolute overlap counts is mathematically flawed because E%-WTA produces smaller assemblies than k-WTA. Orthogonalization claims require normalized metrics (e.g., Jaccard Index).	Data Analysis & Metrics
Reviewer 1 Reviewer 5 PR_2	Trivial Retrieval Testing Testing recovery by presenting the identical, complete original stimulus proves nothing but fixed-point stability. The model must be tested for pattern completion using partial or degraded cues.	Model Capabilities
Reviewer 1 Reviewer 6 Reviewer 8	Missing AC Operations The original AC model relies on specific operations (project, merge, associate). The manuscript fails to test whether the new E%-WTA implementation actually supports these core cognitive functions.	Model Capabilities

Summary Recommendation

To satisfy the reviewers, the core priority for revision should be conducting a **factorial ablation study** separating the selection rule, connection probability, and stopping criteria. Concurrently, updating the **retrieval tasks with partial cues** and **normalizing the overlap metric** will directly address the most

Executive Summary: Reviewers agree that integrating E%-WTA and inhibition into the Assembly Calculus (AC) framework is a valuable pursuit. However, the consensus points to fundamental flaws in experimental design, confounded variables, tautological test definitions, and biologically implausible parameters that undermine the paper's core claims.

Primary Methodological Concerns

Confounded Model Comparison

The comparison between E%-WTA and AC modifies multiple variables simultaneously (selection mechanism, feedforward inhibition, sparsity, and stopping criteria). A factorial ablation is required to isolate the source of the improvements.

- Rev 1
- Rev 3
- Rev 4
- Rev 5
- Rev 6
- Rev 7
- Rev 9

Trivial Retrieval Tests

Testing retrieval by re-presenting the identical, full stimulus to a deterministic network is tautological. To demonstrate true attractor stability and pattern completion, models must be challenged with degraded, noisy, or partial cues.

- Rev 1
- Rev 3
- Rev 5
- Rev 8

Missing Core Operations & Capacity

The study fails to test the fundamental composable operations of the original AC (project, merge), sequence memory, or max storage capacity. Without these, the functional utility of the assemblies remains unverified.

- Rev 1
- Rev 3
- Rev 5
- Rev 6
- Rev 8

Biologically Implausible Sparsity

The selected connection probability ($p_s = 0.5$) is far too high for cortical networks. If the model fails at biologically accurate sparsity levels (0.01 - 0.1), the core claim of biological plausibility is fundamentally compromised.

- Rev 4
- Rev 5
- Rev 9

Technical & Analytical Issues

Shared Critique	Consensus Summary	Reviewers
Unnormalized Overlap	The raw overlap metric used to claim orthogonalization is biased because E%-WTA produces smaller assemblies. A normalized metric (e.g., Jaccard index) is strictly required.	5, 8, 9
Flawed Plasticity Rules	Applying identical, unbounded multiplicative Hebbian learning rules to inhibitory synapses is non-biological. Without synaptic weight normalization, weights will diverge to infinity.	1, 4, 5, 6
Arbitrary Criteria	The new formation criteria mandates the assembly to be a strict fixed point (guaranteeing perfect retrieval by definition) and arbitrarily sets a minimum size of exactly 6 neurons.	3, 5, 7, 9
Flawed Density Metric	Measuring intra-assembly connectivity based purely on the count/presence of connections is inaccurate; it must be quantified by the actual learned connectivity weight strength.	2, 5, 7
Presentation Errors	The phrase "your model" appears repeatedly in the text and captions and must be corrected to "our model".	4, 5, 6, 7, 8, 9
Figure Legibility	Figures suffer from poor resolution, microscopic tick labels, inaccessible color schemes for colorblind readers, and missing error bars on aggregate plots.	2, 4, 8

Note: The combination of the new strict assembly criteria, unnormalized metrics, and the lack of proper degradation tests creates a scenario where the E%-WTA model appears artificially superior to the baseline AC model.

Specific Individual Demands

While the above points represent broad consensus, several reviewers highlighted critical, distinct demands that must be addressed for publication:

Reviewer 2: Mathematical Soundness

Requests analytical results replicating some of the original Assembly Calculus derivations using the new activation function to prove mathematical soundness alongside the simulations.

Reviewer 3: Survival Bias

Points out that reporting statistical medians only for "successful" simulation runs introduces significant survival bias, which skews the overall performance and stability metrics.

Reviewer 6: Dale's Law Violation

Notes that the specific implementation of feedforward inhibition (randomly assigning inhibitory weights across generic synapses) violates Dale's law, further detracting from the model's claim to biological plausibility.

Reviewer 9: Empirical Validation

Emphasizes the need to evaluate the model's output against real in-vivo microcircuit dynamics (e.g., experimental calcium imaging data) to substantiate the lofty claims of biological realism.